causalmetrics lecture note, Section 3

Selection on Observables

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2026-09-05

Contents

1 The problem

1.1 The selection problem in one picture

Let $D \in \{0, 1\}$ be a binary treatment and Y the observed outcome. A naive comparison,

$$\widehat{\Delta}_{\text{naive}} = \bar{Y}_{D=1} - \bar{Y}_{D=0},$$

is almost never causal in an observational setting. Treated and control units typically differ on pre-treatment characteristics, and the naive gap mixes the treatment effect with these covariate differences.

We will work with a single simulated data set throughout Part II so every figure tells a piece of the same story. The data have an imbalanced pre-treatment covariate `x`, a true ATE of 1.5, and a mildly nonlinear outcome surface.

```r
df <- simulate_causal_data(
  n_control = 80,
  n_treated = 60,
  true_ate  = 1.5,
  seed      = 123
)

glimpse(df)
#> Rows: 140
#> Columns: 6
#> $ id    <chr> "C1", "C2", "C3", "C4", "C5", "C6", "C7", "C8", "C9", "C10", "C1~
#> $ treat <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
#> $ x     <dbl> 31.911438, 37.856805, 70.056750, 43.269151, 44.327179, 72.871170~
#> $ group <chr> "Control", "Control", "Control", "Control", "Control", "Control"~
#> $ y0    <dbl> 3.714313, 3.667005, 4.812922, 3.534331, 3.554338, 5.261321, 3.85~
#> $ y     <dbl> 3.714313, 3.667005, 4.812922, 3.534331, 3.554338, 5.261321, 3.85~
```

The naive comparison overstates the true effect because the treated units sit at higher `x`:

```r
naive_diff <- with(df, mean(y[treat == 1]) - mean(y[treat == 0]))
cat("True ATE:           1.50\n",
    "Naive Y difference: ", round(naive_diff, 2), sep = "")
#> True ATE:           1.50
#> Naive Y difference: 2.56
```

```r
plot_opening_scatter(df)
```

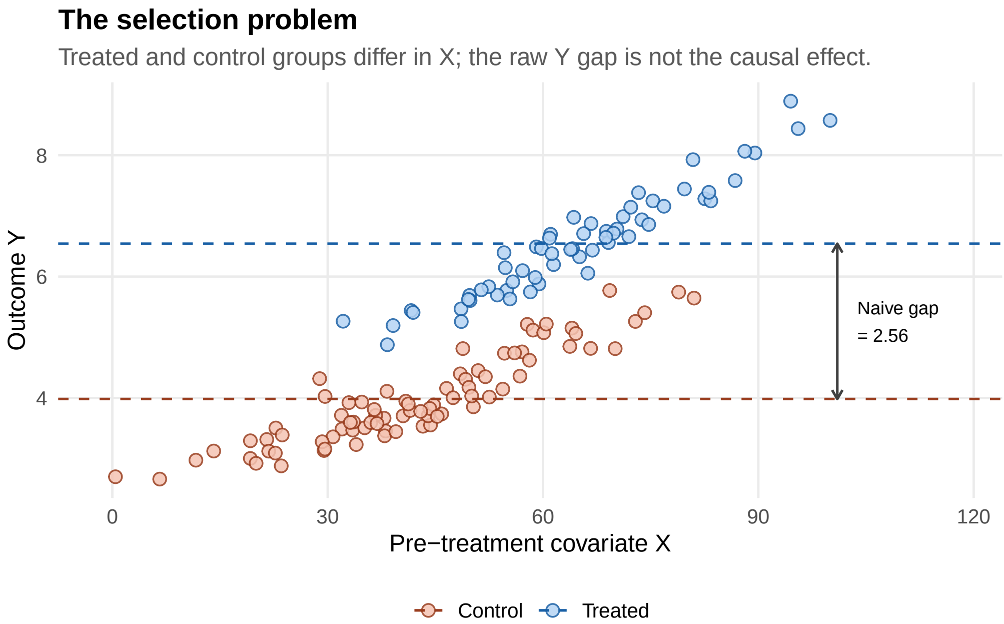

Figure 1: Treated units cluster at higher X, so the raw Y-gap reflects both the treatment effect and the covariate imbalance.

The selection-on-observables design asks whether the pre-treatment covariates we *do* observe are enough to make the comparison credible. If yes, the rest of the toolkit (regression, matching, weighting, AIPW) just makes that comparison in different ways.

1.2 Just enough framework

Potential outcomes. Each unit has two potential outcomes, $Y(1)$ and $Y(0)$. We observe one of them:

$$Y \ = \ D\,Y(1) + (1 - D)\,Y(0).$$

The equality of observed Y with the realized potential outcome is **consistency** — usually quiet, occasionally subtle (it can fail under interference or vague treatment definitions).

Three estimands as design choices. Pick one *before* opening the data:

Estimand	Formula	Natural when
ATE	$\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)]$	Policy applies to everyone
ATT	$\tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid D = 1]$	Question is about participants (job-training, opt-in users)
CATE	$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$	Question is about heterogeneity

The ATE is *not* automatically the right target. If your overlap region is small, the ATT may be both more credibly identified and more managerially meaningful. CATEs and policy learning are the topic of the companion vignette `08_hte_and_policy_learning`.

1.3 Three identifying assumptions, visually

The whole selection-on-observables design rests on three assumptions. Make the case for each *before* showing any estimate.

1.3.1 Conditional ignorability

For the ATE,

$$(Y(1), Y(0)) \perp\!\!\!\perp D \mid X.$$

For the ATT, the weaker one-sided version $Y(0) \perp\!\!\!\perp D \mid X$ suffices.

Plain English: *once we condition on X, treatment assignment is as good as random with respect to the relevant potential outcomes.*

This is a **design** assumption, not a model-fit result. A high predictive R^2, a flexible learner, or a balanced weighted sample cannot prove ignorability. They can only make some failures visible.

1.3.2 Overlap

For the ATE we need $0 < e(X) < 1$ where $e(X) = \mathbb{P}(D = 1 \mid X)$. For the ATT we only need that every treated covariate profile has comparable controls.

Overlap is about *support*. Balance is about *similarity* after the design has been applied. A weighted sample can be balanced only where support exists; without overlap, regression becomes extrapolation and IPW becomes unstable. In our simulated data:

```
plot_ps_overlap(df)
```

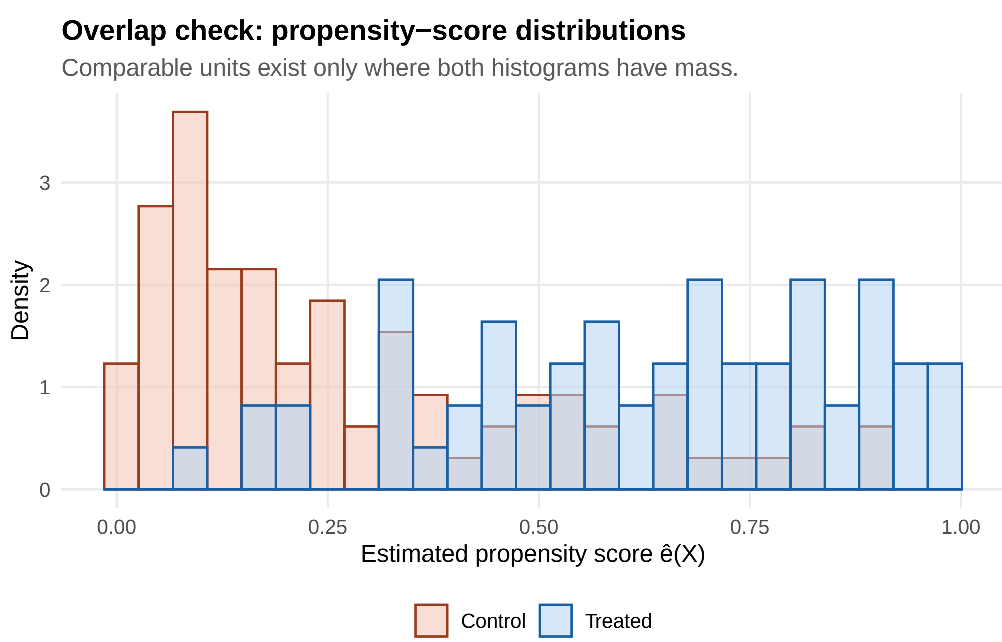

Figure 2: Overlap holds across the bulk of the distribution but thins out at the extremes.

1.3.3 Choosing a valid adjustment set

A safe starting point: adjust for **pre-treatment** common causes of treatment and outcome, justified by institutional knowledge or a DAG. Bad controls can make adjustment worse, not better:

- Post-treatment variables block part of the effect or open mediation paths.
- Conditioning on colliders opens noncausal paths.
- Treatment-only predictors can amplify residual confounding.
- Some pre-treatment variables create M-bias.

The rule is not "control for everything." The rule is *"control for variables that make the causal comparison valid without conditioning on consequences of treatment or colliders."*

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2 Five views of the same problem

Every estimator below either (a) **imputes** the missing potential outcome from a model, (b) finds a **comparable unit** through matching or stratification, (c) **rebalances** the sample by weighting, or (d) **combines** these. The geometry differs; the identifying assumptions do not. Each section follows the same template:

> *Intuition → Picture → Estimator (math) → How to compute it in R → When it works, when it breaks*

2.1 Outcome regression — the imputation view

2.1.1 Intuition

If we knew $\mu_0(x) = \mathbb{E}[Y \mid D = 0, X = x]$, we could *impute* the missing $Y(0)$ for every treated unit and compute the ATT directly. Replace the unknown μ_0 with a fitted $\hat{\mu}_0$, and this becomes a working estimator.

2.1.2 Picture

```
plot_causal_template(df, build_lm_spec(df))
```

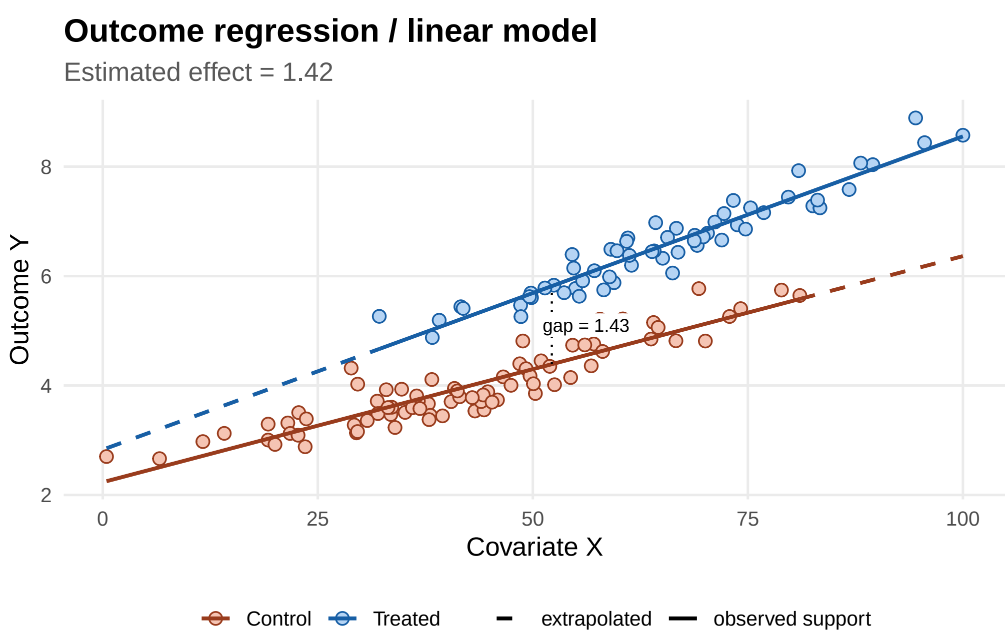


Figure 3: Outcome regression fits separate lines on treated and control. The fitted lines are extrapolated (dashed) into regions where their own group has no data.

The dashed extensions are the source of model dependence: any estimate that averages over the full X distribution depends on the fitted function in regions where one group is unsupported.

2.1.3 Estimator

For the ATT:

$$\hat{\tau}^{\mathrm{RA}}_{\mathrm{ATT}} = \frac{1}{n_1} \sum_{i:\, D_i=1} \{Y_i - \hat{\mu}_0(X_i)\}.$$

For the ATE, fit *both* arms and average the contrast:

$$\hat{\tau}^{\mathrm{RA}}_{\mathrm{ATE}} = \frac{1}{n} \sum_{i=1}^{n} \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

A derivation under one-sided ignorability is in Appendix A.

2.1.4 How to compute it in R

The simplest version is a single regression with treatment as a covariate, using robust standard errors:

```
fit_lm <- estimatr::lm_robust(y ~ treat + x, data = df, se_type = "HC2")
summary(fit_lm)$coefficients["treat", c("Estimate", "Std. Error")]
#>   Estimate Std. Error
#> 1.49550674 0.05931044
```

The "imputation" form is more explicit and lets you target ATT or ATE without forcing a constant treatment effect:

```
# Fit each arm separately, then average the contrast.
fit_c <- lm(y ~ poly(x, 2), data = filter(df, treat == 0))
fit_t <- lm(y ~ poly(x, 2), data = filter(df, treat == 1))

mu0 <- predict(fit_c, newdata = df)
mu1 <- predict(fit_t, newdata = df)

ate_ra <- mean(mu1 - mu0)
att_ra <- mean((mu1 - mu0)[df$treat == 1])

cat("Regression-adjustment estimates:\n",
    "  ATE: ", round(ate_ra, 3), "\n",
    "  ATT: ", round(att_ra, 3), sep = "")
#> Regression-adjustment estimates:
#>   ATE: 1.468
#>   ATT: 1.506
```

2.1.5 When it works, when it breaks

- **Works** when the outcome model is approximately correct on the support of X, and the support overlaps the target population.
- **Breaks** under extrapolation: a smooth, well-fitting model can give apparently precise estimates that depend entirely on a polynomial's behavior outside the data cloud.
- **Symptom of trouble**: results that shift dramatically when you change polynomial degree, interactions, or learner family.

2.2 Matching and subclassification — the find-a-twin view

2.2.1 Intuition

Instead of modeling the outcome surface, find untreated units that look like the treated ones and compare directly. Two variants are common: **propensity-score matching (PSM)** matches on the scalar $\hat{e}(X)$; **coarsened exact matching (CEM)** discretizes X into bins and compares within bins.

2.2.2 Picture: PSM

```
plot_causal_template(df, build_psm_spec(df))
```

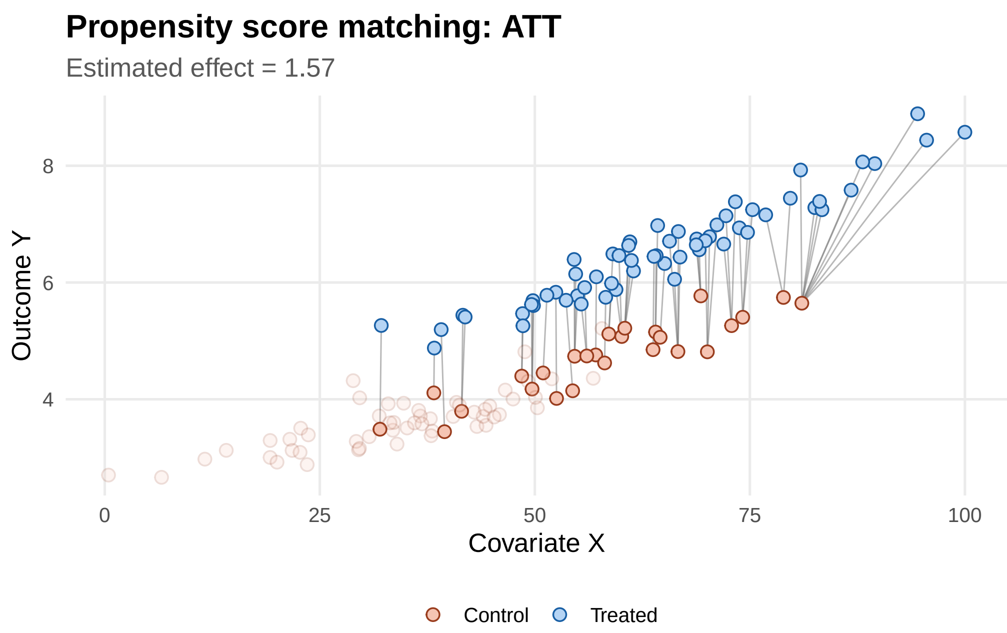


Figure 4: PSM links each treated unit to its nearest control on the propensity score. Unmatched controls (faded) drop out of the comparison.

2.2.3 Picture: CEM

```
plot_causal_template(df, build_cem_spec(df, estimand = "ATT"))
```

2.2.4 Estimator

For PSM (one-to-one, ATT, with replacement):

$$\hat{\tau}_{\text{ATT}}^{\text{PSM}} = \frac{1}{n_1} \sum_{i:\, D_i=1} (Y_i - Y_{j(i)}),$$

where $j(i)$ is the control closest to i on $\hat{e}(X)$.

For CEM, the estimator is a weighted average of within-bin treated-control mean differences; bins where one arm is empty are discarded. This is exactly what `MatchIt::matchit(method = "cem")` returns when you call `match.data()` and run a weighted regression.

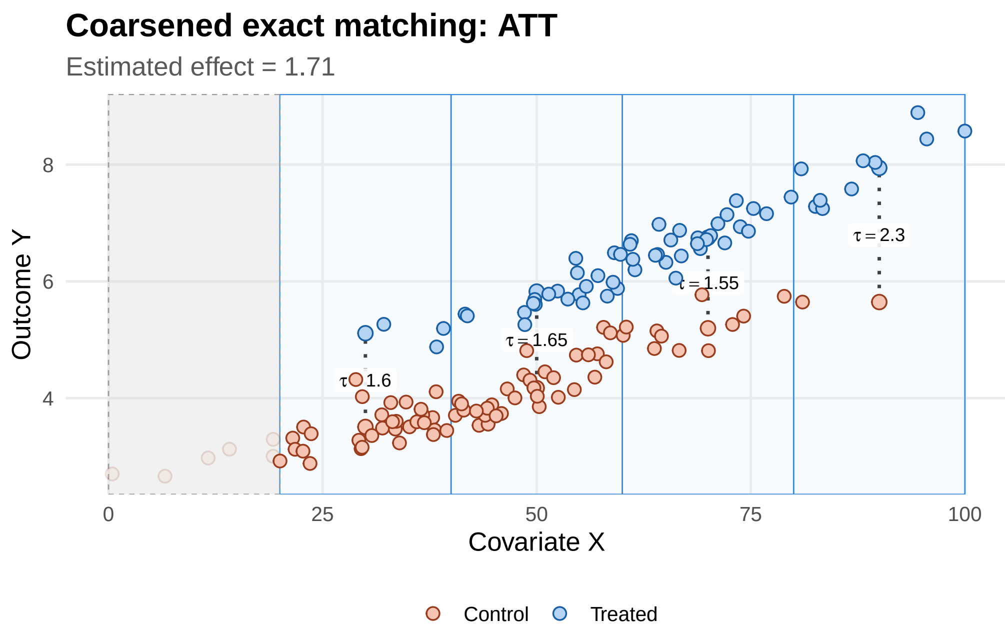


Figure 5: CEM partitions X into bins. Bins with both treated and control units (blue) contribute to the estimate; the rest (grey, dashed) are dropped.

2.2.5 How to compute it in R

`MatchIt` is the modern workhorse. The workflow is *match* → *check balance* → *estimate on the matched sample*:

```
m_psm <- MatchIt::matchit(
  treat ~ x,
  data = df,
  method = "nearest",
  distance = "glm",
  estimand = "ATT",
  replace = TRUE
)

# Inspect what changed:
summary(m_psm)$nn   # sample sizes before and after
#>                Control Treated
#> All (ESS)     80.00000      60
#> All           80.00000      60
#> Matched (ESS) 16.07143      60
#> Matched       26.00000      60
#> Unmatched     54.00000       0
#> Discarded      0.00000       0

# Estimate on the matched sample with robust SEs.
md_psm <- MatchIt::match.data(m_psm)
fit_psm <- estimatr::lm_robust(
  y ~ treat,
  data    = md_psm,
  weights = weights,
  se_type = "HC2"
```

```r
)
summary(fit_psm)$coefficients["treat", c("Estimate", "Std. Error")]
#>   Estimate Std. Error
#>  1.5744660  0.1945526
```

CEM follows the same pattern with `method = "cem"`:

```r
m_cem <- MatchIt::matchit(
  treat ~ x,
  data = df,
  method = "cem",
  estimand = "ATT"
)

md_cem <- MatchIt::match.data(m_cem)
fit_cem <- estimatr::lm_robust(
  y ~ treat,
  data    = md_cem,
  weights = weights,
  se_type = "HC2"
)
summary(fit_cem)$coefficients["treat", c("Estimate", "Std. Error")]
#>   Estimate Std. Error
#>  1.5645681  0.1549215
```

2.2.6 When it works, when it breaks

- **Works** when reasonable matches exist, the estimand is the ATT, and transparency to a non-technical audience matters (you can show "these treated units, these control units").
- **Breaks** as covariate dimension grows: exact matching becomes infeasible; PSM forces all the information into a single dimension and can throw out a lot of variation; CEM bins get sparse.
- **Watch for**: changes in target population. If matching drops many treated units (because no controls look like them), the ATT now refers to a subset.

2.3 Propensity-score weighting — the rebalancing view

2.3.1 Intuition

If treatment is "as good as random" within X levels, we can *re-weight* the sample so that treated and control distributions look identical. The standard weight is the inverse probability of receiving one's own treatment:

$$w_i \;=\; \frac{D_i}{e(X_i)} \;+\; \frac{1-D_i}{1-e(X_i)}.$$

A weighted mean of Y in each arm gives the ATE.

2.3.2 Picture

```r
plot_causal_template(df, build_ipw_spec(df))
```

2.3.3 The weight problem

IPW is fragile under weak overlap. When $\hat{e}(X)$ is close to 0 or 1 for some units, the corresponding weights explode and the estimate is driven by a handful of observations:

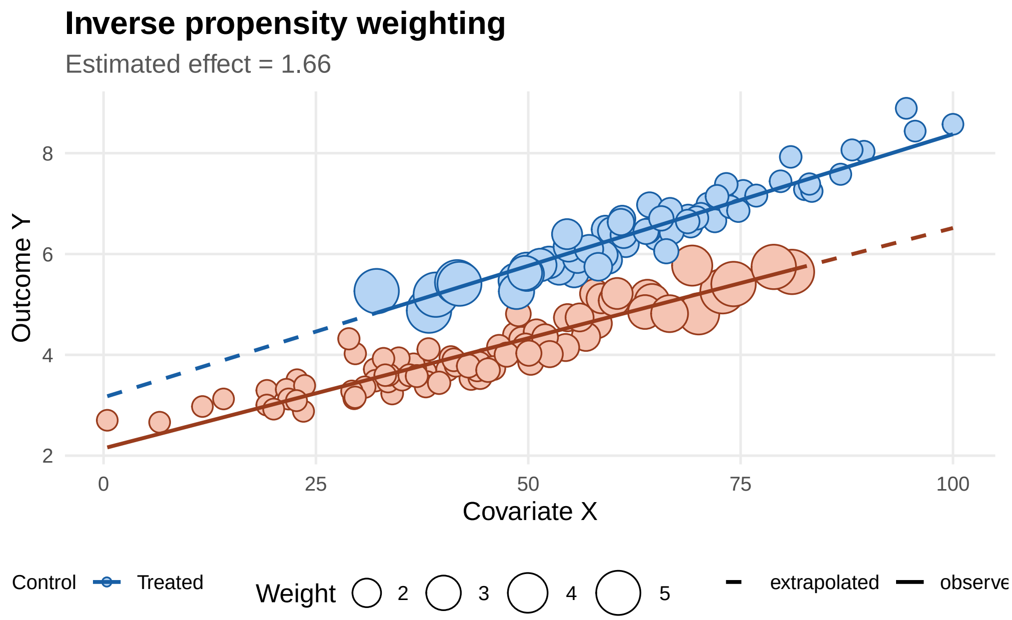


Figure 6: IPW. Point size = inverse-propensity weight. Treated units in low-PS regions and controls in high-PS regions are up-weighted; the weighted regression lines (not the unweighted ones) capture the rebalanced comparison.

```
plot_ipw_weights(df)
```

2.3.4 Estimator

The **Horvitz–Thompson (HT)** form is the textbook expression:

$$\hat{\tau}_{\text{IPW}}^{\text{HT}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{D_i Y_i}{\hat{e}(X_i)} - \frac{(1-D_i)Y_i}{1-\hat{e}(X_i)}\right\}.$$

The **Hájek** form normalizes weights to sum to one in each arm and is almost always more stable in finite samples:

$$\hat{\tau}_{\text{IPW}}^{\text{H}} = \frac{\sum_i D_i Y_i/\hat{e}(X_i)}{\sum_i D_i/\hat{e}(X_i)} - \frac{\sum_i (1-D_i)Y_i/(1-\hat{e}(X_i))}{\sum_i (1-D_i)/(1-\hat{e}(X_i))}.$$

2.3.5 How to compute it in R

```
# 1. Fit the propensity model.
ps_model <- glm(treat ~ x, data = df, family = binomial())

# 2. Compute weights, with light trimming to bound extreme values.
df_ipw <- df %>%
  mutate(
    ps  = pmin(pmax(predict(ps_model, type = "response"), 0.02), 0.98),
    ipw = if_else(treat == 1, 1 / ps, 1 / (1 - ps))
  )

# 3. Hájek-style ATE.
```

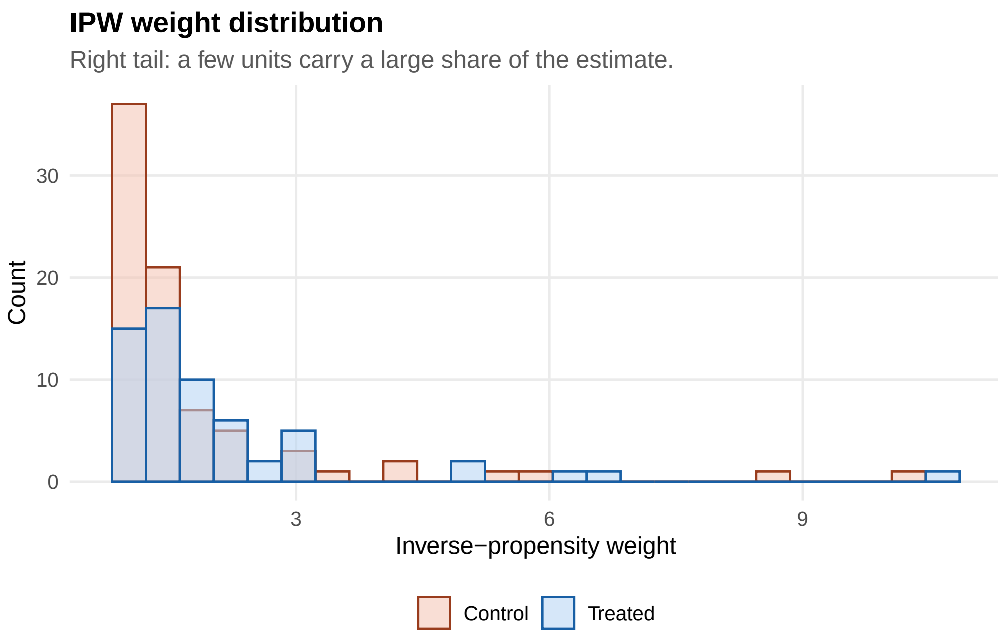


Figure 7: Most weights are modest, but the right tail can contain a few influential units.

```
mu1_hat <- with(df_ipw, sum(treat * ipw * y) / sum(treat * ipw))
mu0_hat <- with(df_ipw, sum((1 - treat) * ipw * y) / sum((1 - treat) * ipw))
ate_ipw_hajek <- mu1_hat - mu0_hat

# 4. Or as a weighted regression with robust SEs (equivalent point
#    estimate, plus inference):
fit_ipw <- estimatr::lm_robust(
  y ~ treat,
  data    = df_ipw,
  weights = ipw,
  se_type = "HC2"
)

cat("Hájek IPW ATE:         ", round(ate_ipw_hajek, 3), "\n",
    "lm_robust(weights):    ",
    round(summary(fit_ipw)$coefficients["treat", "Estimate"], 3),
    sep = "")
#> Hájek IPW ATE:         1.664
#> lm_robust(weights):    1.664
```

2.3.6 When it works, when it breaks

- **Works** when you trust the propensity model more than the outcome model, and overlap is reasonable. The right tool when treatment assignment is well-understood (e.g., randomization within strata, known eligibility rules).
- **Breaks** when propensities approach 0 or 1: a few units carry the estimate, the variance is enormous, and the effective sample size is much smaller than the nominal one.
- **Watch for**: estimates that change a lot when you trim extreme weights (e.g., at the 1st/99th percentile). That is the variance speaking.

2.4 AIPW — combining imputation and rebalancing

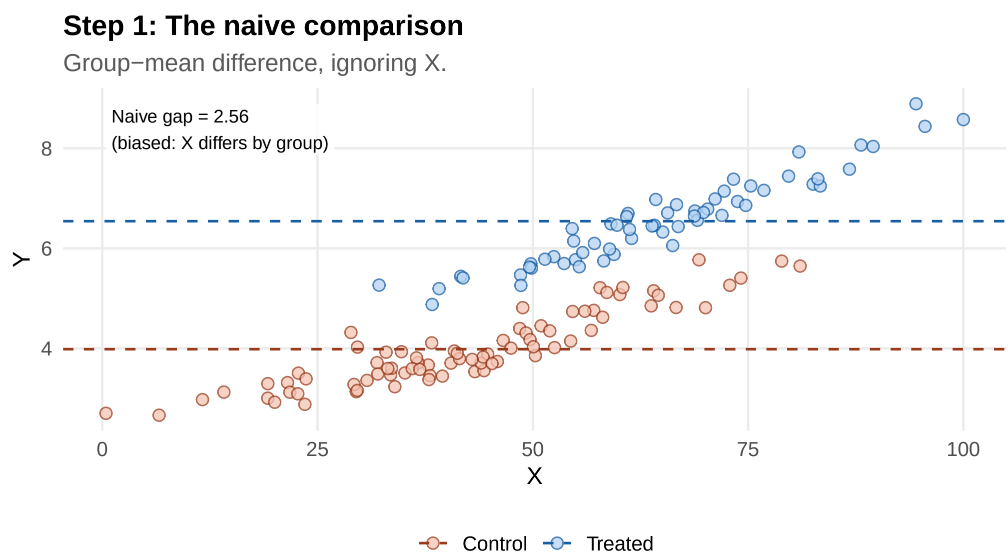
Regression adjustment is model-dependent. IPW is variance-fragile. **Augmented inverse propensity weighting (AIPW)** combines them so that *either* a correct outcome model *or* a correct propensity model is enough for consistency. That is the **double robustness** property.

We tell this story two ways. **Series R** (Robinson's view) is more geometrically intuitive — it's what the partially-linear DML estimator in Section 04 (*Doubly Robust Estimation and Double Machine Learning*) actually computes. **Series A** is the literal AIPW score, which is what `causalmetrics::est_aipw()` runs.

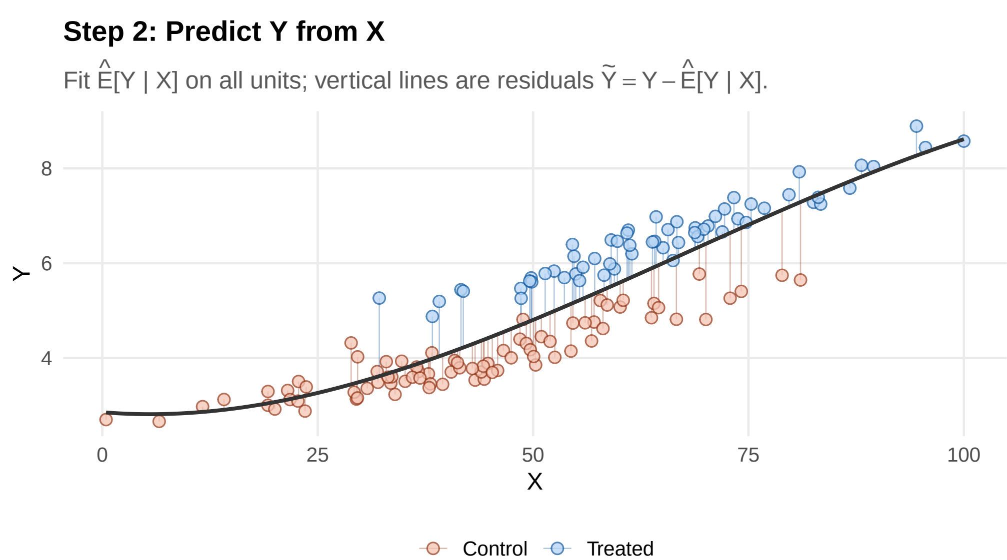
2.4.1 Series R: residualize, then regress

The idea: if X confounds the D–Y relationship, partial X out of *both* D and Y, then regress what is left.

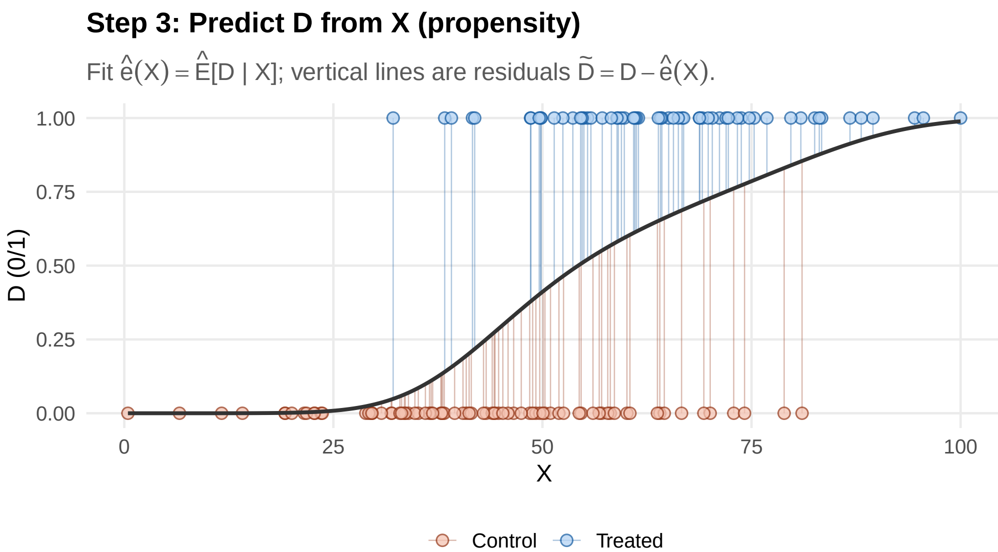
```r
plot_robinson_step1(df)
```



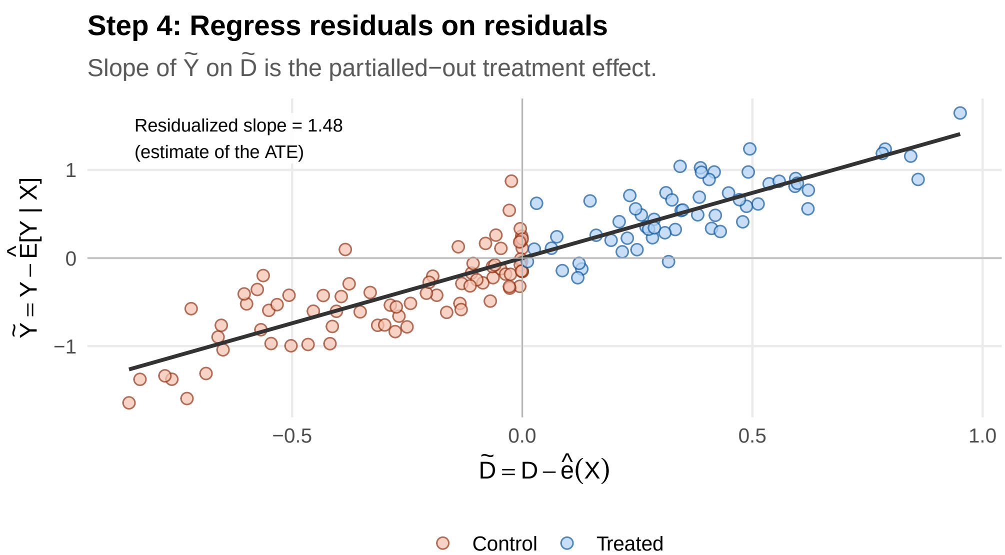

```r
plot_robinson_step2(df)
```




```r
plot_robinson_step3(df)
```




```r
plot_robinson_step4(df)
```



In equations, Robinson's transformation assumes the partially-linear model $Y = \tau D + g(X) + \varepsilon$, defines residuals $\tilde{Y} = Y - \mathbb{E}[Y \mid X]$ and $\tilde{D} = D - e(X)$, and estimates τ by

$$\hat{\tau}_{\mathrm{R}} = \frac{\sum_i \tilde{D}_i \tilde{Y}_i}{\sum_i \tilde{D}_i^2}.$$

This is just an OLS slope of $\tilde{Y}$ on $\tilde{D}$, computed on the *residualized* data. The flexibility lives in how you estimate $\mathbb{E}[Y \mid X]$ and $e(X)$.

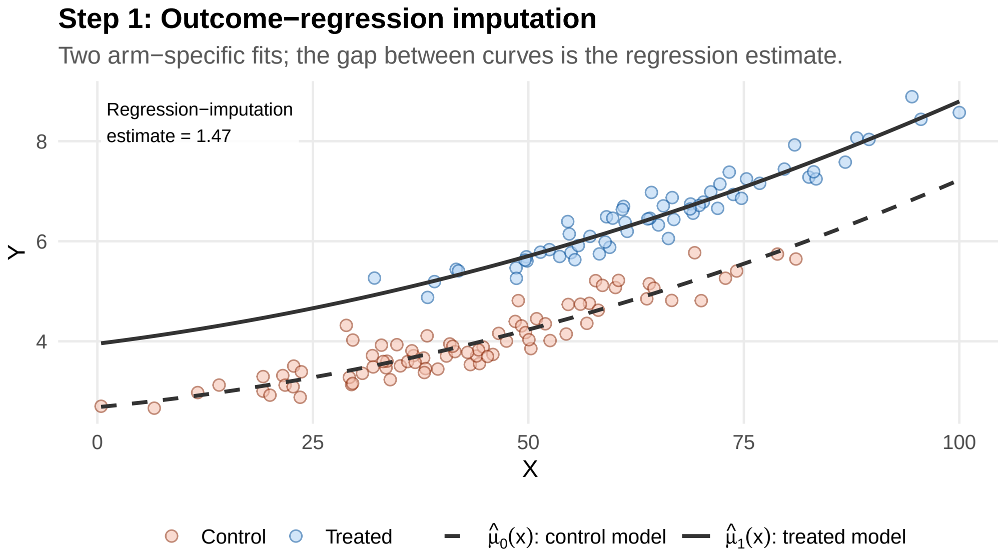
2.4.2 Series A: the classical AIPW correction

The AIPW score is

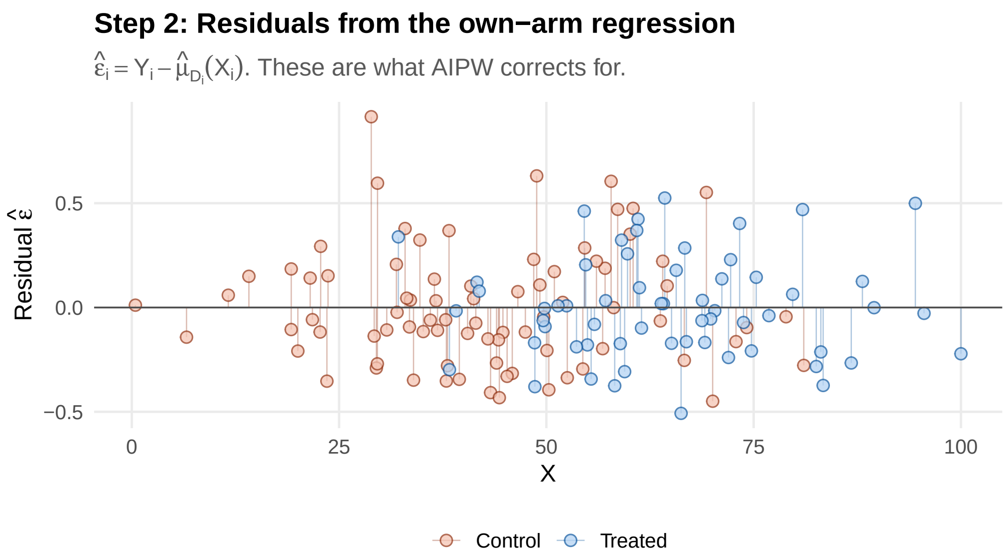
$$\psi_i = \underbrace{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)}_{\text{regression imputation}} + \underbrace{\frac{D_i\{Y_i - \hat{\mu}_1(X_i)\}}{\hat{e}(X_i)} - \frac{(1-D_i)\{Y_i - \hat{\mu}_0(X_i)\}}{1-\hat{e}(X_i)}}_{\text{propensity-weighted residual correction}},$$

and the AIPW estimator is $\hat{\tau}_{\mathrm{AIPW}} = \frac{1}{n}\sum_i \psi_i$.

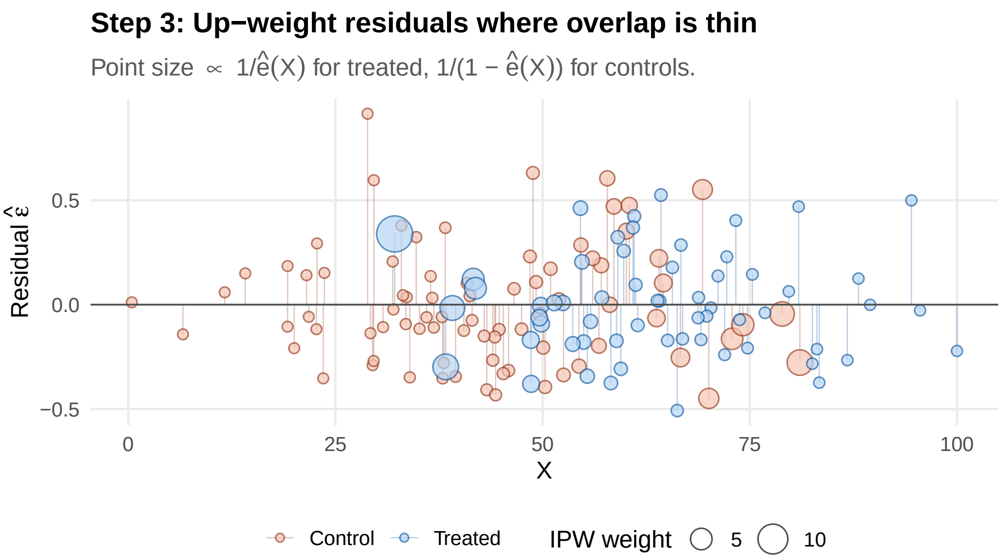
```
plot_aipw_step1(df)
```




```
plot_aipw_step2(df)
```




```
plot_aipw_step3(df)
```




```
plot_aipw_step4(df)
```

Double robustness, plainly. If the outcome model is right, $Y - \hat{\mu}_d(X)$ has mean zero given X, so the correction term contributes nothing in expectation. If the propensity model is right, the weights reweight residuals to debias an incorrect outcome model. Either alone suffices. The Appendix walks through the algebra.

This is robustness *to nuisance-model misspecification*. It is not robustness to hidden confounding, bad controls, missing support, or target-population changes after trimming.

2.4.3 How to compute it in R

`causalmetrics::est_aipw()` evaluates the ATE score directly. It takes a data frame plus column names for the outcome, treatment, and three nuisance predictions $(\hat{e}, \hat{\mu}_0, \hat{\mu}_1)$ — which you can supply from any source: a `glm`, an `mlr3` learner, a Python pipeline, whatever.

```
# Fit nuisance functions (simple polynomials here; in practice
# use cross-fitted ML; see Section 04 on double machine learning).
ps_model <- glm(treat ~ poly(x, 2), data = df, family = binomial())
```

```r
fit_c    <- lm(y ~ poly(x, 2), data = filter(df, treat == 0))
fit_t    <- lm(y ~ poly(x, 2), data = filter(df, treat == 1))

df_nuis <- df %>%
  mutate(
    p_hat   = pmin(pmax(predict(ps_model, newdata = ., type = "response"),
                        0.02), 0.98),
    mu0_hat = predict(fit_c, newdata = .),
    mu1_hat = predict(fit_t, newdata = .)
  )

fit_aipw <- causalmetrics::est_aipw(
  df_nuis,
  y       = "y",
  d       = "treat",
  p_hat   = "p_hat",
  mu0_hat = "mu0_hat",
  mu1_hat = "mu1_hat"
)
fit_aipw
#> AIPW estimate (ATE)
#>   Estimate:   1.4914
#>   Std. Error: 0.0592
#>   95% CI:     [1.3755, 1.6074]
#>   N:          140 (treated: 60, control: 80)
```

Cross-fitting. When nuisance functions are estimated with flexible ML, the same data points should not be used both to fit a nuisance function and to evaluate the score — that re-introduces overfitting bias. Cross-fitting splits the sample into folds and uses out-of-fold predictions in the score. The full DML workflow lives in Section 04 (*Doubly Robust Estimation and Double Machine Learning*); a brief recipe is in Appendix C of this note.

2.4.4 When it works, when it breaks

- **Works** when you want insurance against misspecifying either nuisance model, and you have enough data to estimate both reasonably well.
- **Breaks** under weak overlap: AIPW inherits IPW's instability through the correction term. It also breaks when *both* nuisance models are badly wrong — double robustness is not omni-robustness.
- **Watch for**: large gaps between the regression term and the AIPW estimate. That gap is the correction; if it is huge, the outcome model is probably misspecified somewhere it matters.

2.4.5 A side-by-side recap

```r
plots <- plot_all_estimators(df, cem_estimand = "ATT")
(plots$lm | plots$psm) /
  (plots$ipw | plots$cem) /
  (plots$aipw | patchwork::plot_spacer())
```

When all five estimators land near the same number, you have *estimator-family agreement* — useful but not proof of identification. When they disagree sharply, treat that as a diagnostic: usually overlap is weak, the adjustment set is misspecified, or extrapolation is doing real work.

> **What about meta-learners (S-, T-, X-, R-learner)?** They are the CATE counterparts of the estimators in this section, resting on the same identifying assumptions but targeting

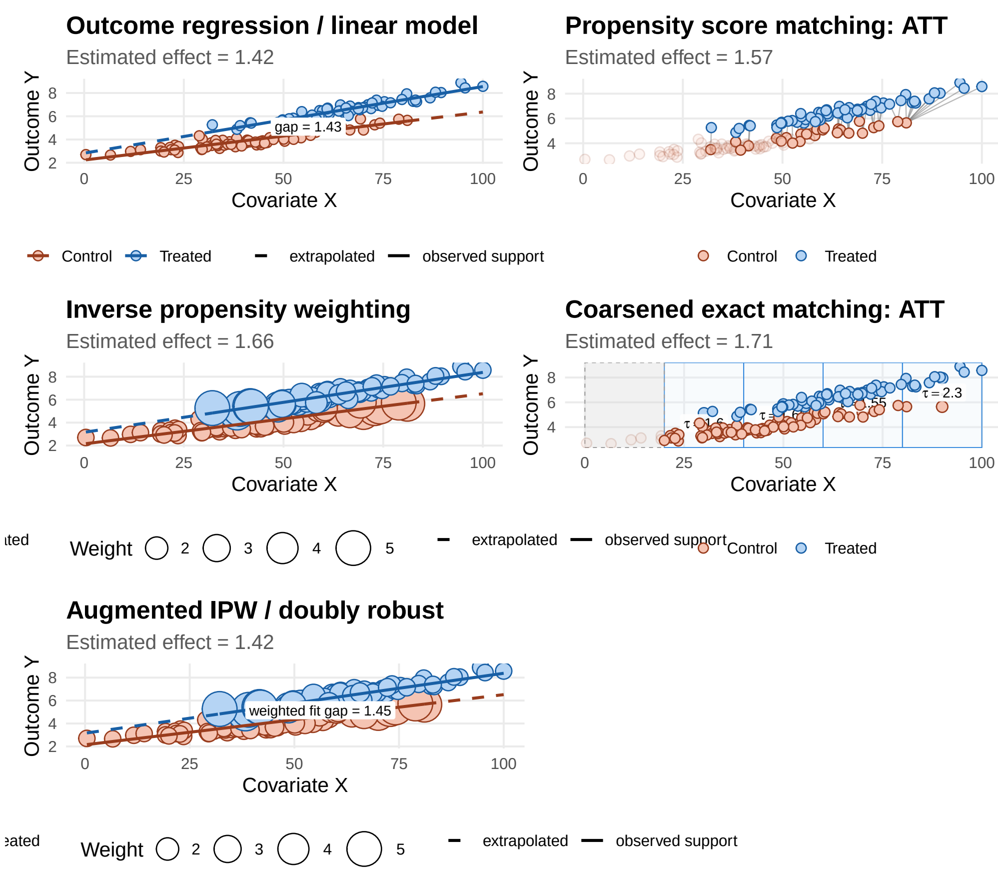


Figure 8: The same data, five estimators. Different geometry, similar destination — when overlap is good and identification holds.

$\tau(x)$ rather than a single number. They get a full treatment, with derivations, in vignette `08_hte_and_policy_learning`.

---

3 Practice

3.1 Choosing among them

There is no universally best estimator. The mapping below is a starting point, not a substitute for thinking about your data:

Situation	Reach for
Strong overlap, smooth outcome surface, low-dimensional X	Outcome regression with robust SEs
Audience wants transparent comparison; ATT is the target	PSM or CEM via `MatchIt`
Assignment process is well-understood; outcome is hard to model	IPW (Hájek form), trim extreme weights
Want insurance against misspecification; enough sample size	AIPW, ideally with cross-fitted ML nuisances
Many high-dimensional controls	AIPW + DML (see vignette 04)
Want (x), not just ¯	Meta-learners (see vignette 08)

A useful sanity-check workflow:

1. Pick the estimand (ATE vs ATT) first; let it constrain everything else.
2. Run **two or three** estimators from different families on the same data.
3. If they agree, report your preferred one and note the agreement.
4. If they disagree, do **not** pick the convenient one. Diagnose overlap, balance, and the adjustment set before continuing.

3.2 Diagnostics in practice

A credible selection-on-observables analysis reports diagnostics *before* interpreting an effect.

3.2.1 Overlap

Already shown: a propensity-score histogram by treatment group (Figure 2). Look for mass in both groups across the propensity range. Trim, restrict to common support, or change the estimand if overlap is weak.

3.2.2 Balance after design

Standardized mean differences (SMDs) on each covariate, before vs. after matching/weighting:

```r
# Prepare three samples to compare.
samples <- list(
  "Raw"        = df %>% select(treat, x),
  "After PSM"  = md_psm %>% select(treat, x),
  "After IPW"  = df_ipw %>% select(treat, x)   # SMD computed unweighted here for simplicity
)
plot_balance(samples, vars = "x")
```

Common rules of thumb: $|\text{SMD}| < 0.1$ is good, 0.1–0.25 is borderline, > 0.25 is poor. These are guidelines, not laws.

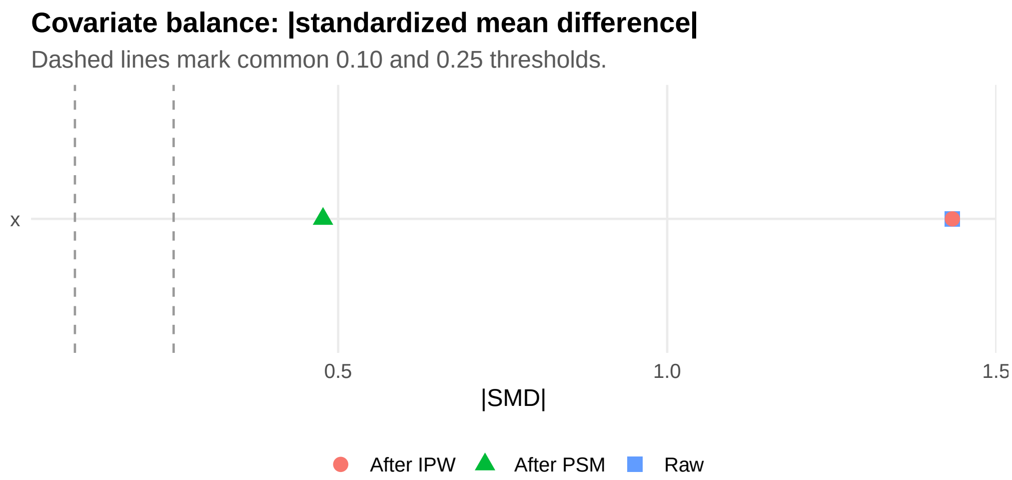


Figure 9: Standardized mean differences before any adjustment, after PSM, and after weighting by IPW.

3.2.3 Weight stability

The IPW weight histogram (Figure 7) is the most direct diagnostic. Report the maximum weight, the share of total weight carried by the top 1% of observations, and any trimming rule.

3.2.4 Estimator-family comparison

Run the same estimand through at least two families. Figure 8 is exactly this diagnostic at one glance.

3.2.5 Placebo and sensitivity

- **Placebo outcomes**: outcomes the treatment *cannot* plausibly affect should yield null effects. A nonzero placebo estimate is evidence against ignorability.
- **Sensitivity analysis** (e.g., Rosenbaum bounds, E-values): asks how strong an unobserved confounder would have to be to overturn the result. Useful precisely because unconfoundedness is untestable.

3.3 Failure modes

The recurring ones, in rough order of how often they bite:

- **No overlap.** Treated units have no comparable controls or vice versa.
- **Hidden confounding.** Observed X does not span the assignment process. No estimator in this family can fix it.
- **Bad controls.** A post-treatment or collider variable in the adjustment set blocks part of the effect or opens noncausal paths.
- **Model dependence.** Estimates shift with functional form, polynomial degree, or learner choice. Often signals weak overlap.
- **Unstable weights.** A handful of observations dominate IPW or AIPW.
- **Target drift.** Trimming and matching change the population the estimate refers to. Report it.
- **A precise estimate of the wrong object.** Modern estimators can agree on the statistical estimand while leaving the *causal* estimand unidentified.

3.4 Replication anchors and checklist

The companion replication in `inst/replications/03_Selection_on_Observables/` uses **Imbens and Xu (2025)**, who revisit LaLonde's job-training benchmark: modern methods can reduce model dependence once overlap is improved, but placebo tests show that observational adjustment may still fail as causal evidence. A second anchor, **Li and Derdenger (2026)**, treats the introduction of NIL rights in college football as a selection-on-observables design and uses AIPW/IPW to compare pre- vs. post-NIL recruits; that replication is planned but not yet part of the package.

A minimal replication checklist:

1. State the causal question and estimand (ATE vs ATT vs CATE).
2. Define $Y(1), Y(0), D, X$ and the observed-outcome relation.
3. Defend the adjustment set as pre-treatment and causally valid.
4. State consistency, ignorability, and overlap explicitly.
5. Check overlap before trusting any adjusted estimate.
6. Estimate nuisance functions flexibly when you have the sample size.
7. Report balance after matching/weighting.
8. Compare estimators from at least two families.
9. Run a placebo and/or formal sensitivity analysis.
10. State how trimming or matching changed the target population.

---

4 Appendices

4.1 Appendix A: Identification derivations

4.1.1 A.1 Regression adjustment for the ATT

Starting from the definition,

$$\tau_{\mathrm{ATT}} \;=\; \mathbb{E}[Y(1)-Y(0)\mid D=1] \;=\; \mathbb{E}[Y\mid D=1]-\mathbb{E}[Y(0)\mid D=1].$$

Apply iterated expectations to the unobserved term:

$$\mathbb{E}[Y(0)\mid D=1] \;=\; \mathbb{E}\big\{\mathbb{E}[Y(0)\mid D=1,X]\;\big|\; D=1\big\}.$$

Under one-sided ignorability $Y(0) \perp\!\!\!\perp D \mid X$ and consistency,

$$\mathbb{E}[Y(0)\mid D=1,X] \;=\; \mathbb{E}[Y(0)\mid D=0,X] \;=\; \mathbb{E}[Y\mid D=0,X] \;=\; \mu_0(X).$$

Therefore,

$$\tau_{\mathrm{ATT}} \;=\; \mathbb{E}[Y\mid D=1]-\mathbb{E}[\mu_0(X)\mid D=1],$$

with sample analogue

$$\hat{\tau}^{\mathrm{RA}}_{\mathrm{ATT}} \;=\; \frac{1}{n_1}\sum_{i:\,D_i=1}\{Y_i-\hat{\mu}_0(X_i)\}.$$

4.1.2 A.2 The AIPW score

Define the score

$$\psi(\eta) \;=\; \mu_1(X)-\mu_0(X) \;+\; \frac{D\{Y-\mu_1(X)\}}{e(X)} \;-\; \frac{(1-D)\{Y-\mu_0(X)\}}{1-e(X)}, \qquad \eta=(\mu_0,\mu_1,e).$$

At the true nuisance values η_0,

$$\mathbb{E}[\psi(\eta_0)] \;=\; \tau_{\text{ATE}}.$$

To see double robustness, write the bias of the score under misspecified $\eta = (\tilde{\mu}_0, \tilde{\mu}_1, \tilde{e})$:

$$\mathbb{E}[\psi(\eta)] - \tau_{\text{ATE}} \;=\; \mathbb{E}\left[\left(1 - \frac{e_0(X)}{\tilde{e}(X)}\right)\{\mu_1^0(X) - \tilde{\mu}_1(X)\} \;-\; \left(1 - \frac{1 - e_0(X)}{1 - \tilde{e}(X)}\right)\{\mu_0^0(X) - \tilde{\mu}_0(X)\}\right],$$

where superscripts indicate truth. The bias vanishes whenever $\tilde{\mu}_d = \mu_d^0$ *or* $\tilde{e} = e_0$. That is the doubly robust property in algebra.

4.2 Appendix B: AIPW influence-function standard error

With $\hat{\psi}_i = \psi_i(\hat{\eta})$ and $\hat{\tau} = n^{-1}\sum_i \hat{\psi}_i$,

$$\widehat{\text{SE}}(\hat{\tau}) \;=\; \sqrt{\frac{1}{n(n-1)}\sum_{i=1}^{n}(\hat{\psi}_i - \hat{\tau})^2}.$$

When nuisance functions come from flexible learners, use cross-fitting or another orthogonalized workflow for valid inference; otherwise the influence-function SE under-states uncertainty.

4.3 Appendix C: Cross-fitting workflow

A minimal design pattern for AIPW with flexible nuisance models:

1. Split the sample into K folds, stratified by treatment.
2. For each fold k: train $\hat{e}$, $\hat{\mu}_0$, $\hat{\mu}_1$ on the *other* $K-1$ folds; predict on fold k.
3. Assemble out-of-fold predictions $(\hat{e}_i, \hat{\mu}_{0,i}, \hat{\mu}_{1,i})$ for every i.
4. Plug into the AIPW score and average.

Cross-fitting reduces sensitivity to first-stage overfitting; it does *not* create overlap, validate the adjustment set, or remove unobserved confounding. The full DML pipeline lives in Section 04 (*Doubly Robust Estimation and Double Machine Learning*).

4.4 References

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